

Objectives

- To create a decentralized printing network within the campus by connecting students who need prints with peers who own personal printers.
- To reduce overcrowding and long waiting times at the central cafeteria printing point.
- To provide students who own printers with an earning opportunity by offering printing services.
- To develop a centralized, role-based online platform for managing print requests, real-time tracking, and secure payments.

Introduction

The proposed platform is a web-based digital solution designed to streamline document printing services within the campus ecosystem. It assists students, teachers, and service providers by making printing easy, fast, and well-organized.

The system uses role-based access, allowing different users—such as students, printer owners, and admins—to access features based on their role. Users can upload documents, select nearby available printers, and track their print requests in real time. By leveraging personal printers already available in student halls, the platform transforms the traditional centralized printing model into a distributed, community-driven network.

Motivation

In our campus (CUET), students face significant problems when they need to print documents urgently—especially during assignment submissions, project deadlines, or thesis presentations. Currently, the only option is the CUET cafeteria, which often becomes overcrowded and time-consuming.

However, many students residing in the halls already have personal printers. If they could register and offer printing services through a dedicated website, other students could print their documents faster and more easily, right from their hall or nearby. This approach will reduce pressure on the cafeteria, save valuable time, and create an opportunity for earning for printer owners.

In brief:

- It will reduce last-minute stress, cafeteria load, and solve the limited printing access problem.
- It will increase community collaboration and provide earning opportunities for students who own printers.
- A centralized online platform will help students easily request print services anytime.

Potential Stakeholders

- **General Students (Users):** The primary consumers who need to print assignments, lab reports, project documents, and thesis papers efficiently without waiting in long queues.
- **Printer Owners (Service Providers):** Students residing in halls who own personal printers and wish to monetize their assets by providing printing services to peers.
- **Administrators:** Individuals responsible for monitoring the system, approving printer registrations, handling payments, and ensuring platform security and integrity.

Existing Related Work

The platform is designed based on real issues faced by CUET students, especially during peak submission periods.

Existing Printing System in CUET:

Currently, CUET has only one main printing point—the cafeteria. During assignment or project submission deadlines, this location becomes overcrowded, resulting in long waiting times and stress for students. Many students have reported difficulties in printing documents on time, especially just before deadlines.

Absence of a Digital Print Platform:

Unlike some international universities that use cloud-based printing systems (e.g., Papercut, Epson Connect), CUET lacks any digital system for managing print requests. Students must physically visit the print point or rely on friends, which is inefficient and unreliable.

Contribution of the Work

This project contributes to the campus environment in several meaningful ways:

- **Stress Reduction:** Significantly reduces last-minute anxiety and physical crowding at the cafeteria printing point by distributing print jobs across the campus.
- **Economic Empowerment:** Provides a structured earning opportunity for students who own printers, fostering financial independence and community collaboration.
- **Improved Accessibility:** Offers a centralized online platform that makes printing services available closer to students' living quarters (halls), unlike the single central location currently available.
- **Digital Transformation:** Moves the campus toward a more connected environment where services are requested and managed digitally with real-time tracking and secure payment options.

Proposed Features of the Web Application

The proposed features of the Print Automation platform include:

1. Home Page

- Action buttons: "Add Printer", "Request Print"
- About Us section emphasizing community engagement
- User testimonials and reviews
- Authentication checks for all interactive actions

2. Authentication

- JWT-based login/signup system
- Role-based access control (User, Printer Owner, Admin)
- Profile dropdown for role-switching for Admin and Printer Owners

3. User Dashboard

- User-specific metrics and recent activity feeds
- Option to "Apply as Printer Owner"
- Comprehensive printer registration form
- Request print form with active printer listings
- Quick-action panels: credit management and print requests

4. Printer Owner Dashboard

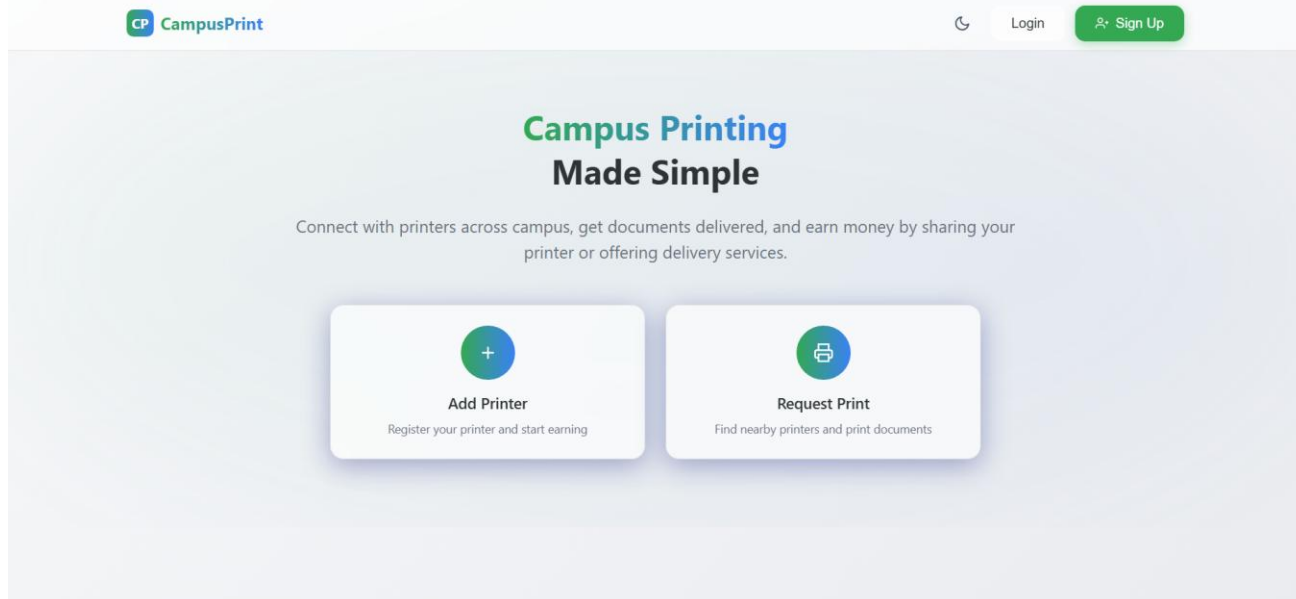
- Management of printer status: active, offline, busy
- Grid layout displaying print requests with accept/reject functionality
- Financial dashboard for monitoring income and handling withdrawal requests (Bkash)

5. Admin Dashboard

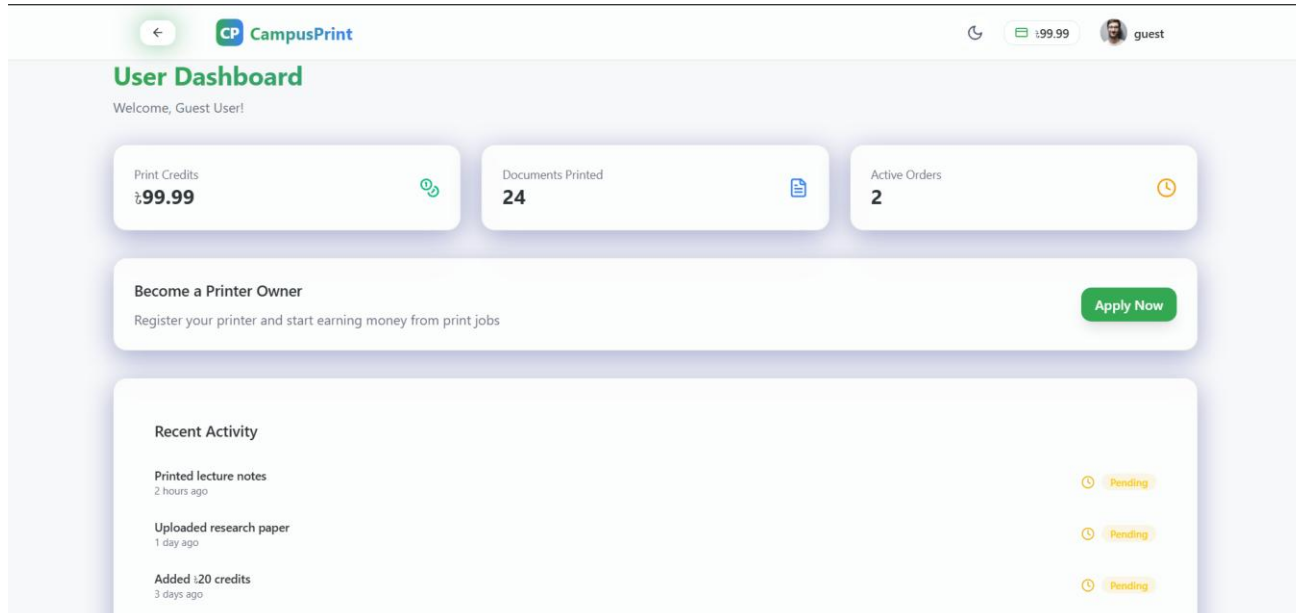
- User management: detailed views, modification, banning users
- Printer approval and verification
- Analytics dashboard for monitoring system performance (user growth, request completion, transaction logs)
- System database reset functionality for historical data cleanup

Expected UI Design

1. Frontend



2. Student's Dashboard



3. Print Owner Dashboard

CP CampusPrint

₹99.99

Demo Printer Owner

Printer Owner Dashboard

Welcome, Guest Printer Owner!

Monthly Revenue

₹342.50

Print Jobs

89

Active Printers

3

Recent Activity

Job completed - Biology notes

30 minutes ago

Pending

New print request

1 hour ago

Pending

Printer maintenance completed

2 hours ago

Pending

Quick Actions

4. Admin's Dashboard

CP CampusPrint

₹99.99

guest

Admin Dashboard

Welcome, Guest User!

Total Users

1,234

Active Printers

45

Monthly Revenue

₹12,450

Recent Activity

New printer registered

1 hour ago

Pending

User report resolved

2 hours ago

Pending

System maintenance completed

1 day ago

Pending

Quick Actions